

Name: _____

Class: _____

Date: _____

Lab 6: Binary Converter

Python Lab · Section 2.1 Binary Numbers



Goal

Use Python to convert between binary and decimal, and explore bits, overflow, and roundoff.

Warm-up: Predict

```
print(bin(10))
print(int("1010", 2))
print(2 ** 8)
```

Predict each line. What does 0b mean?

AP Connection

- n bits can represent 2^n values (2.1).
- Overflow and roundoff errors (2.1).

Build It

Step 1. Convert decimal to binary with `bin()`. Remove the 0b with slicing.

```
n = int(input("Decimal number: "))
print(bin(n)[2:])
```

Step 2. Convert binary to decimal with `int(text, 2)`.

```
b = input("Binary number: ")
print(int(b, 2))
```

Step 3. How many values can n bits hold? Print a table for 1 to 10 bits.

```
for bits in range(1, 11):
    print(bits, "bits:", 2 ** bits, "values")
```

Step 4. Simulate 8-bit overflow: an 8-bit number can only hold 0–255.

```
x = 250
x = (x + 10) % 256
print(x) # what happened?
```

Step 5. Roundoff: try this and explain the result.

```
print(0.1 + 0.2)
print(0.1 + 0.2 == 0.3)
```

Date:

Build It (continued) · Level Up

Explain in your own words why $0.1 + 0.2$ isn't exactly 0.3 on a computer.